

Application Development 1 Project Report

Student Details

Name: Kriti Tiwari

Roll Number: 24F2005768

Email: 24f2005768@ds.study.iitm.ac.in

About Me: I am a student at IIT Madras BS in Data Science and Programming. I am currently at diploma level. I am interested in arts and reading. I wish to be able to make meaningful contributions to the field of data science.

Project Details

Project Title: Hospital Management System

Project Statement: To design and build a web-based application that allows Admins, Doctors, and Patients to interact with the system based on their roles.

Approach: The app was built using Flask as the backend framework for hospitals, doctors and patients to easily manage appointments and treatments.

AI/ LLM Declaration

No use of AI/ LLM.

Technologies and Frameworks Used

1. **Flask:** Core backend web framework
2. **Flask-Login:** User authentication and session management
3. **Jinja2:** Template engine for rendering dynamic HTML pages
4. **SQLite:** Lightweight local database for storing user data
5. **SQLAlchemy:** Object Relational Mapper for SQLite Database
6. **Datetime:** Convert string to date type object and store into the database
7. **Dateutil:** Relative difference between two dates
8. **Pyisemail:** Validating user's email
9. **Bcrypt:** Storing hashed passwords in database
10. **Regex re:** Backend Validation for HTML forms
11. **Bootstrap 5 Icons:** Icons

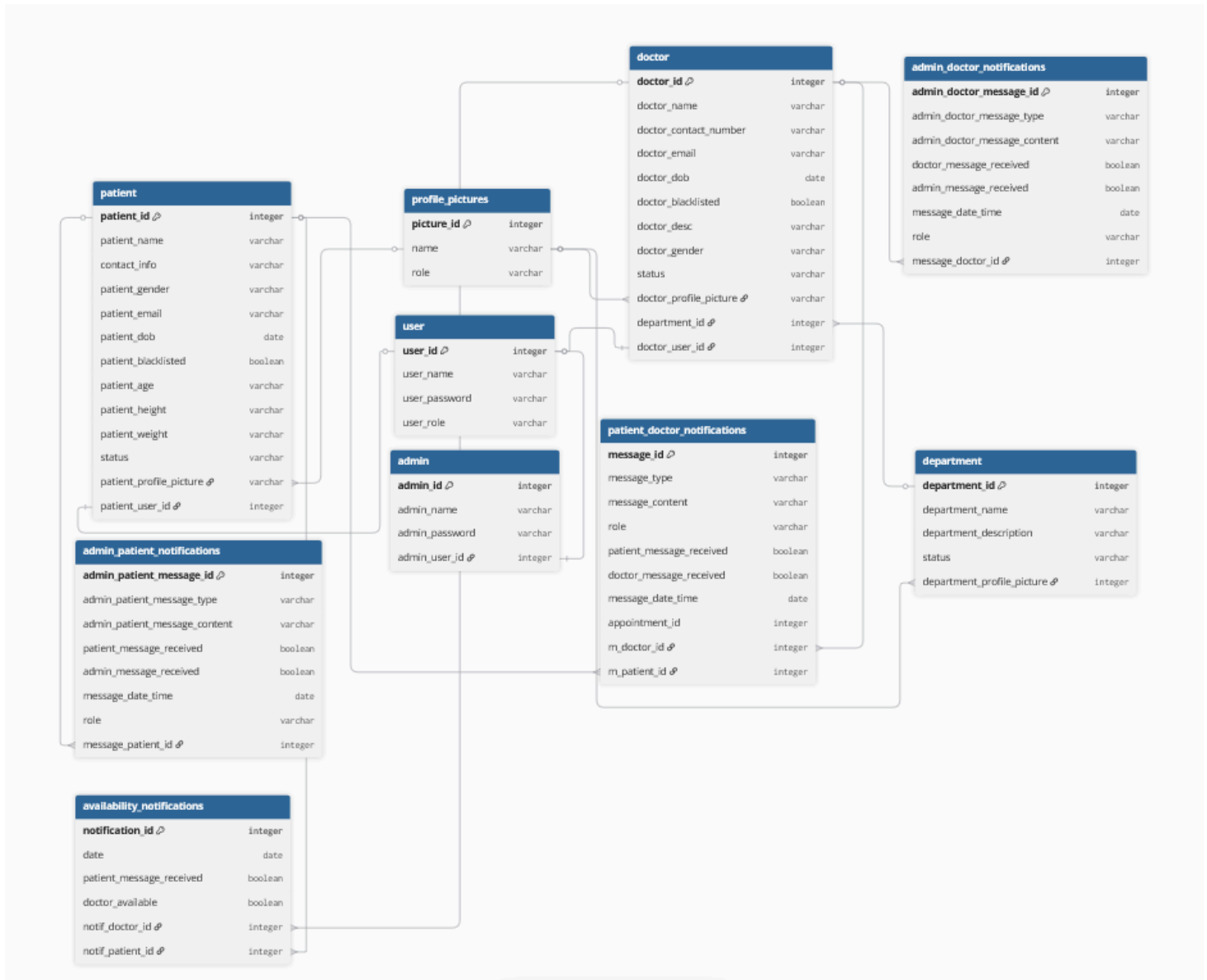
Database Schema/ ER Diagram

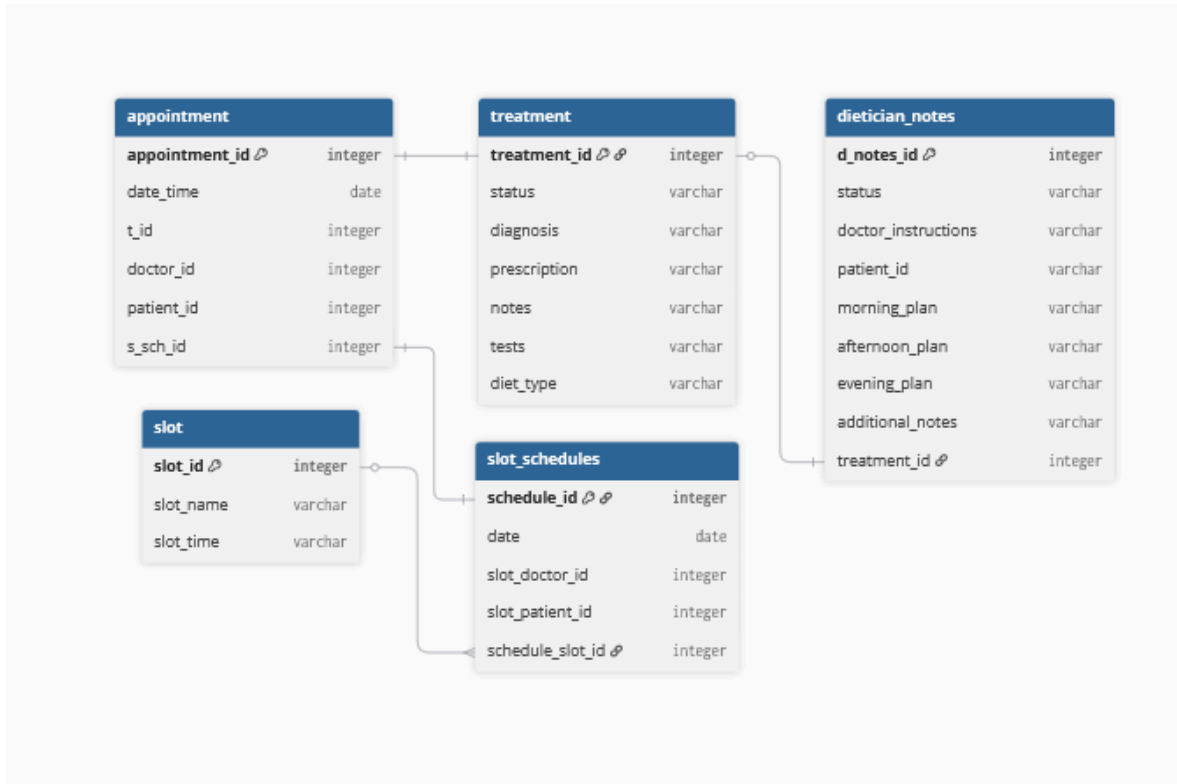
Tables:

1. **User:** stores user name, password and their role.
2. **Admin:** stores admin data.
3. **Department:** stores department name, description and status.
4. **Doctor:** stores doctor name, contact number, email, date of birth, description, gender, status, profile picture and the blacklisted status
5. **Patient:** stores patient name, contact number, gender, date of birth, height, weight, status, profile picture and the blacklisted status.
6. **Appointment:** stores data about appointments date time, doctor ID, patient ID and slot ID
7. **Treatment:** stores diagnosis, tests, prescription and notes provided by the doctor for an appointment.
8. **Slot:** stores data about slot name and time.
9. **SlotSchedules:** logs data about doctor appointments with patients such as slot timing and patient IDs.
10. **AvailabilityNotifications:** Waiting list for the patients who chose to be notified about a doctor's availability.
11. **PatientDoctorNotifications:** stores messages between patients and doctors.
12. **AdminPatientNotifications:** stores messages between admin and patients.
13. **AdminDoctorNotifications:** stores messages between admin and doctors.
14. **DieticianNotes:** stores the diet plans given by the hospital dietetician.
15. **ProfilePictures:** stores the name, role and image name of images used throughout the app.

Relationships

- **One-to-One relationships:** User-Admin, User-Doctor, User-Patient, Appointment-Treatment, Treatment-DieticianNotes, Appointment-SlotSchedules
- **One-to-Many relationships:** ProfilePictures-Department, ProfilePictures-Doctor, ProfilePictures-Patients, Slot-SlotSchedules, Patient-AvailabilityNotifications, Doctor-AvailabilityNotifications, Doctor-PatientDoctorNotifications, Patient-PatientDoctorNotifications, Doctor-AdminDoctorNotifications, Patient-AdminPatientNotifications





Architecture and Features

Architecture Overview:

- [app.py](#) - main Flask application entry point
- /application/[models.py](#) - database models using SQLAlchemy
- /application/[controllers.py](#) - Flask functions for doctors, patients and admin
- /create_app.py - initialises and creates the database
- /templates - Jinja2 HTML templates
- /static/images - Images used for the Flask application
- /static/style.css - CSS used for the Flask application
- /static/[script.js](#) - JavaScript used for the Flask application

Implemented Features

- Admin can perform the following functionalities:
 - Add, update and delete Patient profiles

- Add, update and delete Doctor profiles
- Add, update and delete Departments
- Search for patients, doctors and departments by their name, contact numbers or IDs
- View all appointments
- Blacklist patients and doctors
- Doctor can perform the following functionalities
 - View all assigned appointments
 - Mark appointments as completed or cancelled
 - Enter diagnosis, tests, notes and prescriptions for each appointment.
 - View complete medical history of the patients
 - Update and provide their availability for the next 7 days
 - Search for other doctors, all departments and the patients assigned to the doctor
- Patients can perform the following functionalities:
 - Register, login and update their profiles
 - View and search doctors by specialization or name
 - View, book, reschedule, or cancel appointments.
 - View their own appointment and treatment history.

Additional Features:

- Patients can receive personalized diet plans by the hospital's dietetician.
- Notifications: Users can send and receive messages to communicate on events like appointment rescheduling or cancellation.

Video Presentation

https://drive.google.com/file/d/1M1_-qjrZvaJus0lpr1y5GGp6rSJg8SPH/view?usp=sharing